Difference between Fibre and Sclereids

Fibre	Sclereids
• Long cells • Narrow, Elongated pointed ends • Occurs in bundles • Commonly unbranched • Derived directly from meristematic tissue	• Short cells • Usually short and broad • Occurs individually or in small groups • Maybe branched • Develops from secondary sclerosis of parenchyma cells

Difference between Tracheids and Fibres

Tracheids	Fibres
• Not much elongated • Possess oblique end walls • Cell walls are not as thick as Fibres • Possess various types of thickenings • Responsible for the conduction and also mechanical support	• Very long cells • Possess tapering end walls • Cell wall are thick and lignified • Possess only pitted thickenings • Provide only mechanical support

Difference Between Sieve Cells and Sieve Tubes

Sieve cells	Sieve tubes
• Have no companion cells • The sieve areas do not form sieve plates • The sieve areas are not well differentiated • They are elongated cells and are quite long with tapering end walls • The sieve are smaller and numerous • Found in Pteridophytes and Gymnosperms	• Have companion cells • The sieve areas are confined to sieve plates • The sieve areas are well differentiated • They consist of vertical cells placed one above the other forming long tubes connected at the walls by sieve pores • The sieve pores are longer and fewer • Found in Angiosperms

9.3 The Tissue System

Introduction to Tissue System, Types and Characteristics of tissue System

As you have learnt, the plant cells are organised into tissues, in turn the tissues are organised into organs. Different organs in a plant show differences in their internal structure. This part of chapter deals with the different type of internal structure of various plant organs and its adaptations to diverse environments.

A group of tissues performing a similar function, irrespective of its position in the plant body, is called a ***tissue system***. In 1875, German Scientist ***Julius Von Sachs*** recognized three tissue systems in the plants. They are:

1. Epidermal tissue system (derived from protoderm)
2. Ground tissue system (derived from ground meristem)
3. Vascular tissue system (derived from procambium)



Histology

(Greek. histos – web, logos – science) It is the study of tissues, their composition, and structure as observed with the help of microscope.

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